

Dhruv Agrawal

Bengaluru, India | [LinkedIn](#) | +91-9528176750 | dhruvagrwal1055@gmail.com

WORK EXPERIENCE

Transunion

Bengaluru

Associate Software Developer

June 2024-present

- Architected a modular C++ execution engine with a plugin-based architecture, enabling realtime loading of region-specific transformation logic and ML and Non-ML models across 10+ regions, and leveraging Apache Arrow-based columnar data pipelines for efficient cross-language processing.
- Collaborated with backend and platform teams to integrate the execution engine with Ray Serve and gRPC-based services, contributing to scalable distributed inference and achieving ~16 ms latency at ~300 requests/sec.
- Enabled distributed computing by integrating Vertex AI Ray clusters alongside Dataproc, improving large-scale data processing performance by 45%, and coordinating with data engineering teams for seamless adoption.
- Led the design and development of a migration and provisioning workflow for user-created Jupyter notebooks, enabling seamless transition to Vertex AI managed notebooks while maintaining backward compatibility, improving deployment consistency and increasing automation efficiency by 60%.
- Established Role-Based Access Control (RBAC) across multiple services, reducing unauthorized access incidents to zero in production and strengthening access governance across the platform.
- Documented system architecture and workflows across multiple services, improving cross-team understanding and reducing onboarding time for new contributors.
- Took ownership of key system components, collaborating with cross-functional teams to drive architectural decisions and performance improvements.

PROJECTS

Workforce Management SaaS

Aug 2025-Oct 2025

Java, Spring Boot, PostgreSQL, Kafka, Redis

- Engineered a multi-tenant SaaS platform supporting concurrent users for workforce scheduling and attendance, reducing manual scheduling effort by 40%.
- Designed and developed microservices for user management, scheduling, and notifications, enabling scalable, modular architecture for concurrent workforce workflows.
- Implemented authentication and role-based authorization using Spring Security and JWT, securing access across multiple user roles and service endpoints.
- Integrated Kafka for event-driven communication and Redis caching to handle high-frequency operations, reducing response times under concurrent load by 35%.

Notes Application

Mar 2023-Jun 2023

Kotlin, Jetpack Compose, Room Database, Firebase

- Built an Android note-taking application using Kotlin and Jetpack Compose with MVVM architecture, supporting offline-first storage and Firebase-based synchronization, reducing crashes by 40%.
- Designed responsive UI state management using Coroutines and LiveData to improve user experience and application reliability.

EDUCATION

Lovely Professional University

Phagwara, Punjab

Bachelor of Technology (B.Tech), Computer Science and Engineering

Aug 2020-May 2024

- CGPA:- 8.58

SKILLS

Languages: Java | C++ | Python | JavaScript | Kotlin

Frameworks: Spring Boot | FastAPI | Node.js | Express.js | React.js | Jetpack Compose

Distributed Systems: Apache Arrow | ONNX Runtime | Ray Serve | gRPC | Dataproc

Databases: MySQL | MongoDB | Room | Firebase (Auth, Firestore, Cloud Messaging)

Cloud & Tools: Google Cloud Platform | AWS | CI/CD (Harness, GitHub Actions, Cloud Build) | Git/GitHub | Docker

Architecture: Microservices | Plugin-Based Architecture | Zero-Copy Data Transfer | Columnar Storage | Distributed Systems | REST APIs